Priyanka Natarajan

NLP Research Engineer

priyanka.natarajan@example.com · Chennai, India

SUMMARY

NLP engineer with six years of experience in text classification, retrieval and large language model fine-tuning. I turn research papers into production services and measure everything with proper offline and online evaluation. I publish internal technical write-ups and mentor interns on experiment design and reproducibility. I am equally comfortable reading papers and reading production logs, and I care about reproducible experiments.

SKILLS

PyTorch, Transformers, Hugging Face, spaCy, Text classification, Embeddings, Vector search, LoRA fine-tuning, Python, scikit-learn, MLflow, Weights & Biases, FAISS

EXPERIENCE

NLP Research Engineer, Verbatim Cloud Labs 2021 – Present

- Fine-tuned a multilingual encoder for support ticket routing, lifting macro F1 from 0.71 to 0.86, and wrote the documentation that new team members use to reproduce the benchmark.
- Built an embedding based retrieval service for knowledge base answers serving 90 queries per second, which allowed the product team to ship the feature without adding GPU capacity.
- Set up evaluation harnesses with human-labelled test sets and drift monitoring.
- Ran LoRA experiments for domain adaptation of a 7B parameter model.
- Reduced inference cost by 55% through distillation of a large classifier into a compact student model.

Data Scientist, Tamarind Fintech 2018 – 2021

- Trained fraud models with gradient boosting and calibrated thresholds with the risk team.
- Prototyped named entity extraction for KYC documents.
- Built an end-to-end text analytics prototype and presented it to executives, which led to a funded product team.
- Cleaned and labelled 40,000 documents with domain experts.
- Introduced experiment tracking and model cards, giving risk and product teams a shared view of performance.

Research Intern, Madras Language Technologies 2017 – 2018

- Experimented with word embeddings and sequence models for sentiment analysis.
- Wrote evaluation scripts and reported results weekly.
- Contributed to a conference paper on low-resource language tagging.
- Created data augmentation scripts that improved recall on rare document types.
- Worked with annotators and reviewers to refine labelling guidelines.

EDUCATION

- MTech Artificial Intelligence, Coromandel Institute of Technology (2016 – 2018)

LANGUAGES

English (fluent), Tamil (native)